

# Nguyen Tran Duc Hoang

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## Summary

Final-year Computer Science student at HCMUT specializing in **Computer Vision**, with **AI Engineer internship experience at CT Group – Gascolae in object detection and multi-object tracking**. Hands-on across the full model pipeline, from data preparation to deployment. Seeking an **AI Engineer (Computer Vision) Fresher** position.

## Education

**HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY (HCMUT)** Ho Chi Minh City, Vietnam  
*Bachelor of Computer Science. GPA: 3.2/4.0* Expected 2027  
Relevant Coursework: Machine Learning, Data Mining, Programming for AI and Data Science, Data Structures and Algorithms, Database Systems.

## Experience

**CT GROUP – GASCOLAE** Ho Chi Minh City, Vietnam  
*AI Engineer Intern* Aug 2026 – Oct 2026

- Worked on object detection and multi-object tracking for computer vision applications in the company's AI team.
- Took part in the model development workflow, from data preparation and training to evaluation and iteration.

## Projects

**Smart Traffic Monitoring: Vehicle Detection, Tracking and Counting** | [GitHub](#) Jul 2026 – Sep 2026  
*YOLO11m, Ultralytics, PyTorch, ByteTrack, OpenCV, Streamlit*

- Fine-tuned YOLO11m on BDD100K (70K training / 10K validation images, 1.28M boxes, 10 classes) on two Tesla T4 GPUs with mixed precision; selected the best of 4 checkpoints at mAP@50 0.58 / mAP@50–95 0.33 on validation, with consistent test results (0.44 mAP@50–95 on the 4 vehicle classes).
- Built a data validation pipeline (corrupt-image scan, image–label pairing, YOLO label checks) that repaired 9.2K invalid boxes.
- Implemented ByteTrack/BoT-SORT tracking with persistent IDs and a directional line-crossing counter; delivered a Streamlit app that exports annotated H.264 video and per-event CSV reports.

**Facial Emotion Recognition Web Application** | [GitHub](#) Sep 2025 – Dec 2025  
*TensorFlow, Keras, ResNet101, OpenCV, Flask*

- Fine-tuned ResNet101 on RAF-DB (7 emotions) with two-stage transfer learning, class weighting and data augmentation, reaching 70.6% test accuracy; deployed it in Flask with OpenCV face detection.

**IoT Smart Home Access Control** | [GitHub](#) Jan 2026 – Jun 2026  
*Face++ API*

- Integrated the Face++ API for face recognition into a smart-home system for access control and user verification.
- Implemented camera input processing and decision logic to connect recognition results with device control.
- Integrated IoT communication components to support real-time monitoring and automated access actions.

**Credit Risk Prediction** | [GitHub](#) Mar 2026 – Apr 2026  
*XGBoost, Scikit-learn, SHAP, Streamlit*

- Built an XGBoost default-risk model on 600K+ Lending Club loans with class weighting and threshold tuning (ROC-AUC 0.742 vs. 0.716 baseline), explained with SHAP and served in a Streamlit dashboard.

## Skills

**Computer Vision:** Detection (YOLO11, Ultralytics), multi-object tracking (ByteTrack), ResNet, OpenCV  
**Machine Learning:** PyTorch, Scikit-learn, XGBoost, SHAP, NumPy, Pandas, Matplotlib  
**Programming & Tools:** Python, C++, SQL, Streamlit, Flask, REST APIs, Docker, Git  
**Language:** English (TOEIC Listening & Reading 610)